Project Overview

Build a deterministic immune simulation sandbox with agents, molecule fields, and chemotaxis. Focus on performance, determinism, and modular design.

Milestone 0: Setup

Create Unity project structure. Implement SimulationController with fixed timestep and pause/step functionality.

Milestone 1: Spatial Grid

Implement spatial hashing grid. Support insert, update, and neighbor queries. Avoid allocations using pooling.

Milestone 2: Cell System

Create data-oriented cell storage. Update positions and age. Avoid MonoBehaviour per agent.

Milestone 3: Molecule Field

Implement grid-based concentration field with diffusion and decay. Add lesion emitters.

Milestone 4: Chemotaxis

Compute gradients and move agents toward higher concentration. Blend with randomness.

Milestone 5: Phenotypes

Add M0/M1/M2 types with different lifespans and behaviors.

Milestone 6: Determinism

Seed RNG, enforce fixed update order, verify identical runs.

Milestone 7: Performance

Add profiling, optimize memory usage, remove allocations.

Milestone 8: Visualization

Render agents and heatmaps. Use clear, accessible colors.

Milestone 9: Config System

Load simulation parameters from JSON/XML.

Milestone 10: C++ Plugin (Optional)

Offload heavy computation like diffusion or neighbor search.

Milestone 11: Portfolio Prep

Prepare README, demo video, and performance results.